

CodeNova Business Website

Testing & QA Document

Version: 1.0 **Project:** CodeNova Business Website

Defines the quality-assurance strategy and release test coverage for the CodeNova website, admin dashboard, forms, security, responsiveness, accessibility, SEO and performance.

1. QA Objectives

- Verify functional correctness.
- Verify responsive behavior.
- Prevent security regressions.
- Ensure forms and lead flows work end-to-end.
- Verify SEO and accessibility requirements.
- Ensure production build quality.

2. Functional Test Cases

- Home page loads successfully.
- Navigation links open correct pages.
- Service cards open correct service pages.
- Desktop Application Development page is present and correct.
- Portfolio filters work.
- Case studies open correctly.
- Blog listing/article pages work.
- Contact form accepts valid data.
- Invalid contact data shows validation.
- Estimator calculates an indicative range.
- Consultation form submits successfully.
- WhatsApp CTA opens the intended chat flow.

3. Admin Test Cases

- Admin login succeeds with valid credentials.
- Invalid credentials are rejected.
- Unauthorized users cannot access admin routes.
- Lead list loads.
- Lead search/filter works.
- Lead status can be changed by authorized users.
- Lead notes/assignment work.
- Portfolio CRUD works.
- Blog draft/publish flow works.

- Service management works.
- Admin logout invalidates the session.

4. API Tests

- Validation rejects missing required fields.
- Protected endpoints reject unauthenticated requests.
- Protected endpoints reject unauthorized roles.
- Rate limits activate under abuse thresholds.
- Correct HTTP status codes are returned.
- Malformed IDs are handled safely.
- Database failures produce safe errors.

5. Security Tests

- Check XSS payloads in all user-input fields.
- Check SQL injection attempts.
- Check CSRF protections where applicable.
- Check insecure direct object reference attempts.
- Check admin authorization bypass attempts.
- Verify secrets are absent from frontend bundles.
- Verify sensitive errors are not returned to clients.
- Verify secure cookies/headers where applicable.

6. Responsive Tests

- Test 360px mobile.
- Test 390px mobile.
- Test 768px tablet.
- Test 1024px tablet/small desktop.
- Test 1280px desktop.
- Test 1440px desktop.
- Test 1920px large desktop.
- Verify no horizontal scrolling and no broken layouts.

7. Browser Tests

- Current Chrome.
- Current Edge.
- Current Firefox.
- Current Safari where available.
- Mobile Chrome.
- Mobile Safari.

8. Accessibility QA

- Keyboard-only navigation.
- Visible focus states.
- Form labels and error announcements.
- Heading hierarchy.
- Color contrast.
- Alt text.
- Reduced motion.
- No critical accessibility blockers.

9. SEO QA

- Unique page titles.
- Unique meta descriptions.
- Canonical URLs.
- Sitemap valid.
- Robots configuration valid.
- Open Graph metadata.
- Structured data validated where used.
- No broken internal links.
- Correct HTTP status codes.

10. Performance QA

- Lighthouse audit.
- Core Web Vitals review.
- Image optimization.
- Check layout shifts.
- Check JavaScript payloads.
- Test slow network conditions.
- Verify third-party scripts do not unnecessarily block rendering.

11. UAT / Acceptance

- Business owner reviews copy and branding.
- All services are accurately represented.
- Portfolio claims are verified.
- Contact details are correct.
- Pricing/estimates are clearly labeled as applicable.
- Privacy/security links are present.
- Production domain and analytics are verified.

12. Release Gate

- No critical or high-severity unresolved defects.
- Critical user journeys pass.
- Security checks pass.
- Production build succeeds.
- Database migrations succeed.
- Monitoring is active.
- Rollback plan is documented.

Document Relationship

This document complements the CodeNova PRD, TRD, Security & Privacy Document and Frontend Requirements Document. Implementation should use the approved business information and actual production configuration rather than placeholders or unverified claims.